

ER diagram: course registration system

Set up:

1. Feel free to do this exercise individually or as a team.
2. Follow the instructions below and document your results in a copy of this document or using a diagramming tool of your choice (feel free to extend the diagram on page 2, or draw your diagram in any way you want.)
3. **Submit your proposed solution here:** <https://forms.gle/S6oxD2PWCqX4b6HN8>

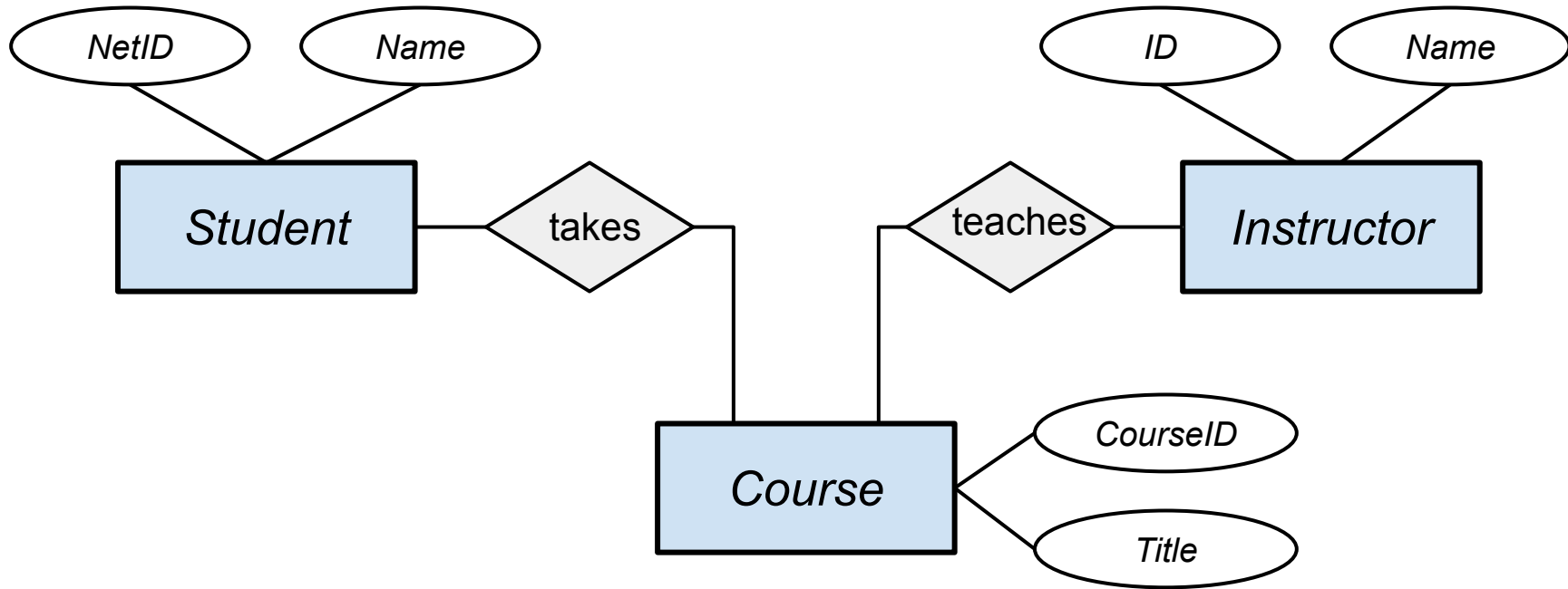
Instructions:

1. Augment the simple ER diagram on the next slide by
 - a. **modelling** the following concepts:
 - i. **Overall grade for a taken course**
 - ii. **Course prerequisites**
 - iii. **Assignments (and points)**
 - b. **adding keys and cardinalities**
2. Think about:
 - a. Which concepts should be modeled as (weak) entities vs. relationships vs. attributes?
 - b. Are there any self-references? What are reasonable cardinalities (reflecting reality)?

Additional Information:

- ER syntax and rules are summarized on the remaining slides

ER diagram: course registration system



ER diagram syntax and rules

ER diagrams: overview

- An Entity Relationship (ER) diagram is a **graphical representation** of a **data model**.
- It shows the **relationship** between **entities** (e.g., people, objects, events, or concepts) within a system.
- It **can be mapped** to a **relational** (database) **schema**.

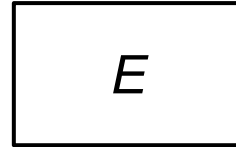
ER diagrams: graphical syntax

- An entity *E*

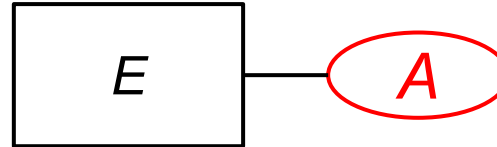


ER diagrams: graphical syntax

- An entity E

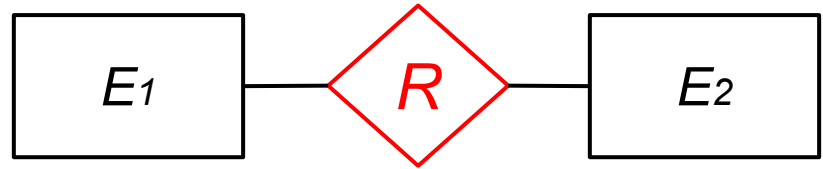
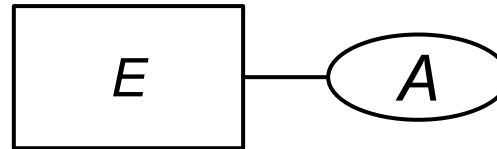
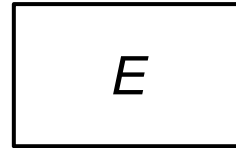


- An attribute A of entity E



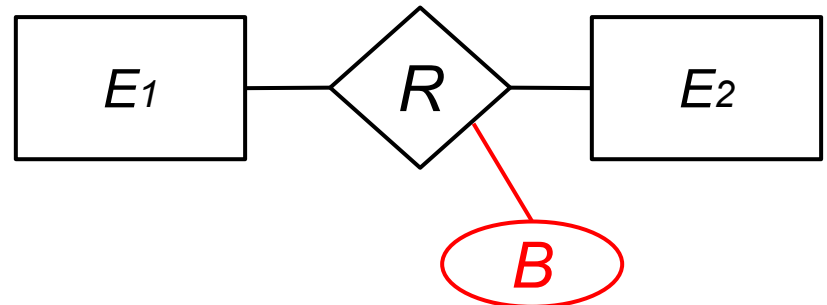
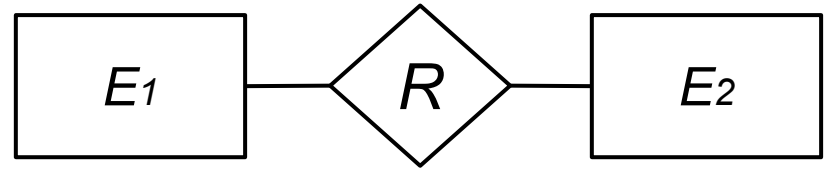
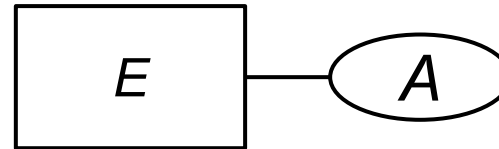
ER diagrams: graphical syntax

- An entity E
- An attribute A of entity E
- A **relationship** R between two entities E_1 and E_2



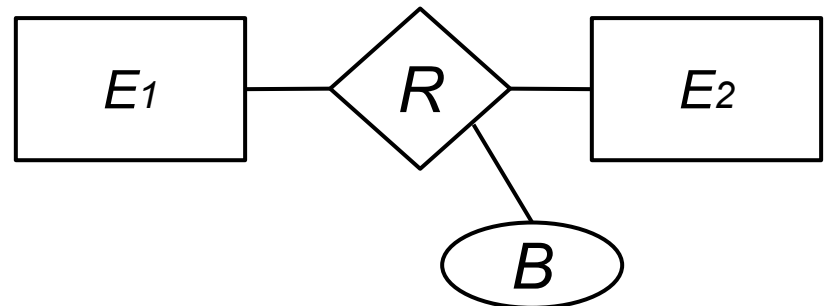
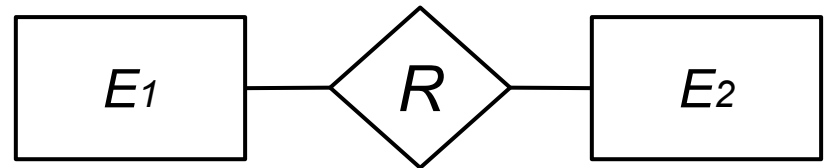
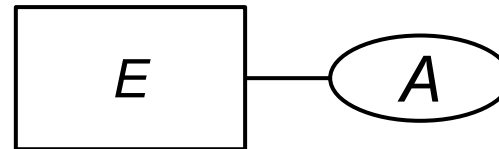
ER diagrams: graphical syntax

- An entity E
- An attribute A of entity E
- A relationship R between two entities E_1 and E_2
- An **attribute B** of relationship **R**



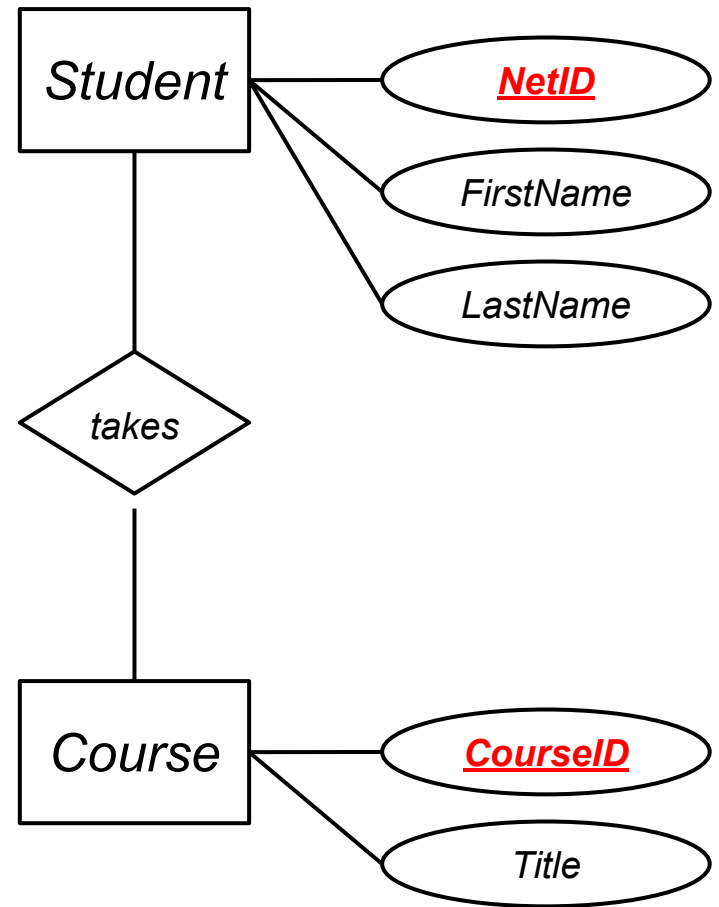
ER diagrams: rules

- An interconnecting line is only allowed between:
 - a box and a diamond,
 - a box and an oval,
 - a diamond and an oval.
- An oval must have exactly one connecting line.
- Names of boxes must be unique in the diagram.
- Names of ovals must be unique per box/diamond.



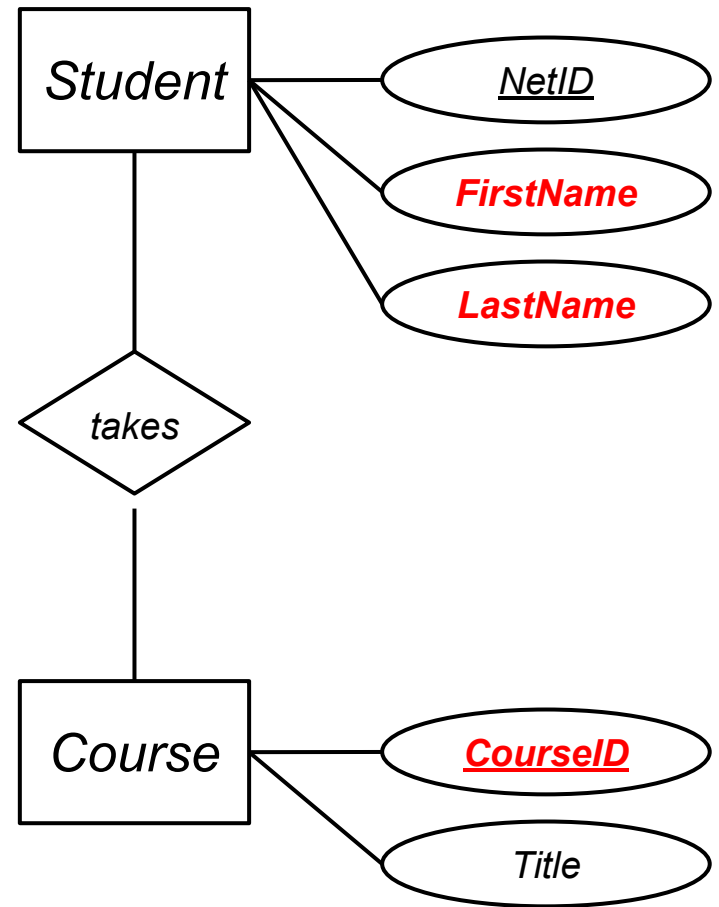
ER diagrams: keys and cardinalities

- A **key** is an (underlined) attribute, or a set of attributes, which **uniquely identifies an entity**.



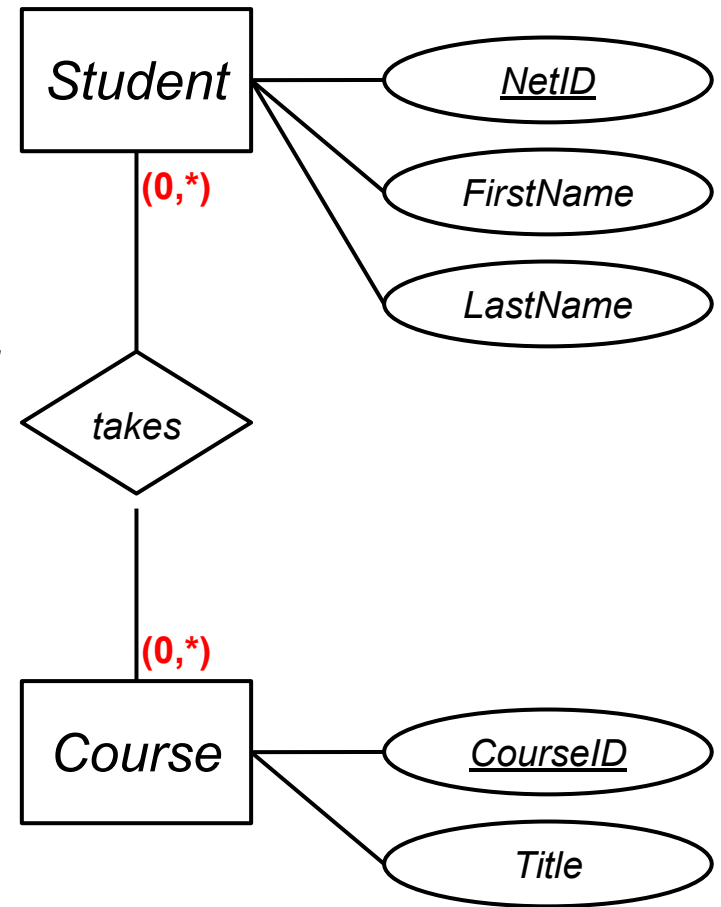
ER diagrams: keys and cardinalities

- A key is an (underlined) attribute, or a set of attributes, which uniquely identifies an entity.
- A key can be **artificial** or **natural**.



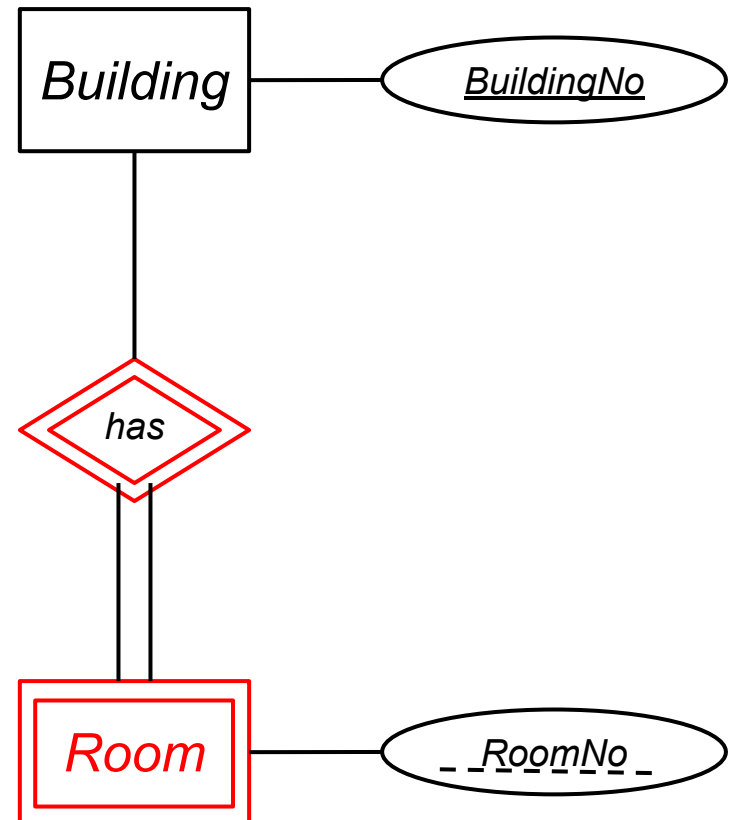
ER diagrams: keys and cardinalities

- A key is an (underlined) attribute, or a set of attributes, which uniquely identifies an entity.
- A key can be artificial or natural.
- The **cardinalities** define the kind of relationship (**one-to-one**, **one-to-many**, or **many-to-many**).
- There are different notations for cardinalities. For example:
 - 1 = (1,1)
 - c = (0,1)
 - m = (1,*)
 - mc = (0,*)



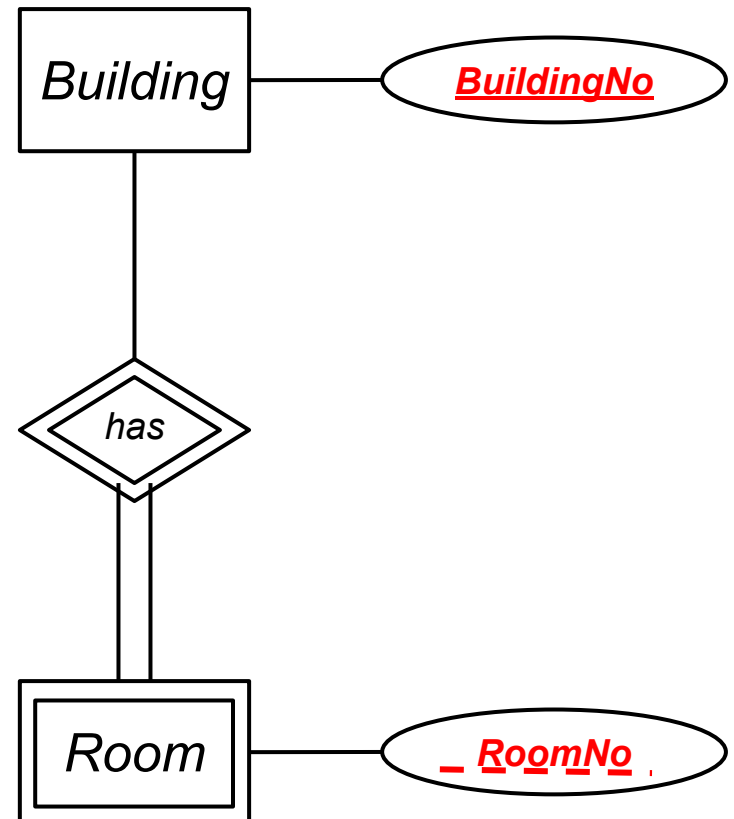
ER diagrams: weak entities

- A **weak entity** can't exist on its own (if a building is torn down, its rooms disappear).



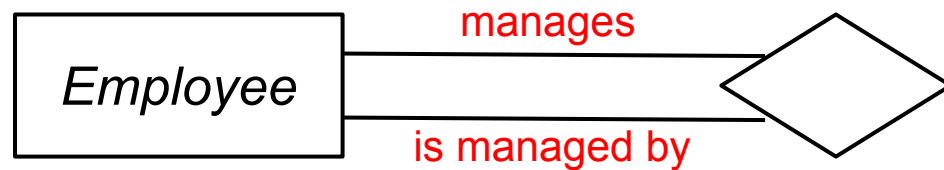
ER diagrams: weak entities

- A weak entity can't exist on its own (if a building is torn down, its rooms disappear).
- A weak entity is only **uniquely identifiable** in **reference** to another entity.



ER diagrams: self references and roles

- A **self reference** is usually explicitly annotated with **roles** to clarify the meaning of the self-referencing relationship.



Think about (but never draw) the following:

